

# Project 22: Movie Revenue Prediction

## Phase 3: Final Solution and Report

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### 1 Introduction and Objectives

The objective of this project is to predict global movie box office revenue. Following the midterm evaluation, the model was significantly upgraded to fulfill the requirements of the final phase. We transitioned from utilizing a restricted data subset to processing the entire TMDB Box Office Prediction dataset to capture a wider distribution of variance and improve predictive robustness.

### 2 Description of the Implemented Solution

The feature space was expanded and refined. Through feature engineering, we extracted temporal signals (`release_year`, `release_month`) from datetime strings to compensate for missing attributes present in other TMDB subsets (such as `vote_average` and `vote_count`). Furthermore, we integrated audience perception metrics (`popularity`) alongside the core financial (`budget`) and logistical (`runtime`, `original_language`) features, resulting in a robust 6-feature pipeline.

To maximize predictive performance, we upgraded from linear baselines to an **XGBoost Regressor** (Extreme Gradient Boosting). This architecture builds decision trees sequentially to correct residual errors, making it highly effective at capturing non-linear relationships and mitigating the impact of heavy-tailed outliers found in blockbuster movie revenues. The final model utilizes 200 estimators with a learning rate of 0.05 and a maximum depth of 6.

### 3 Instructions to Run the Code

The complete final solution is version-controlled and fully contained within the provided Python script (`project.py`). To replicate the final results:

1. Ensure Python 3.9+ is installed along with the required dependencies: `pandas`, `numpy`, `scikit-learn`, `matplotlib`, `seaborn`, and `xgboost`.
2. Download the full `train.csv` from the official TMDB Box Office Prediction Kaggle competition and place it in the root directory alongside the script.
3. Execute the script via terminal: `python project.py`.
4. The script will autonomously preprocess the full dataset, train the XGBoost model, print the final evaluation metrics to the console, and generate a visualization (`final_results_plot.jpg`) in the working directory. Reproducibility is guaranteed via a globally initialized `SEED = 42`.

## 4 Results and Analysis

Model performance was evaluated using Root Mean Squared Error (RMSE), the Coefficient of Determination ( $R^2$ ), and Mean Absolute Percentage Error (MAPE). MAPE was specifically introduced in this final phase to provide a scale-independent understanding of prediction accuracy.

The formal definitions of our core evaluation metrics are as follows:

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (1)$$

$$\text{MAPE} = \frac{100}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right| \quad (2)$$

Our final optimized XGBoost model yielded the following metrics:

- **Final  $R^2$  Score:** 0.6352 (*Improved from Midterm baseline of 0.5655*)
- **Final RMSE:** \$95,393,568.79
- **Final MAPE:** 2096.76%

Transitioning to the full dataset and implementing the Extreme Gradient Boosting architecture successfully increased our explained variance ( $R^2 \approx 0.64$ ). The inclusion of temporal signals and the popularity metric proved to be highly correlated with financial success, allowing the decision trees to better separate blockbuster hits from standard releases.

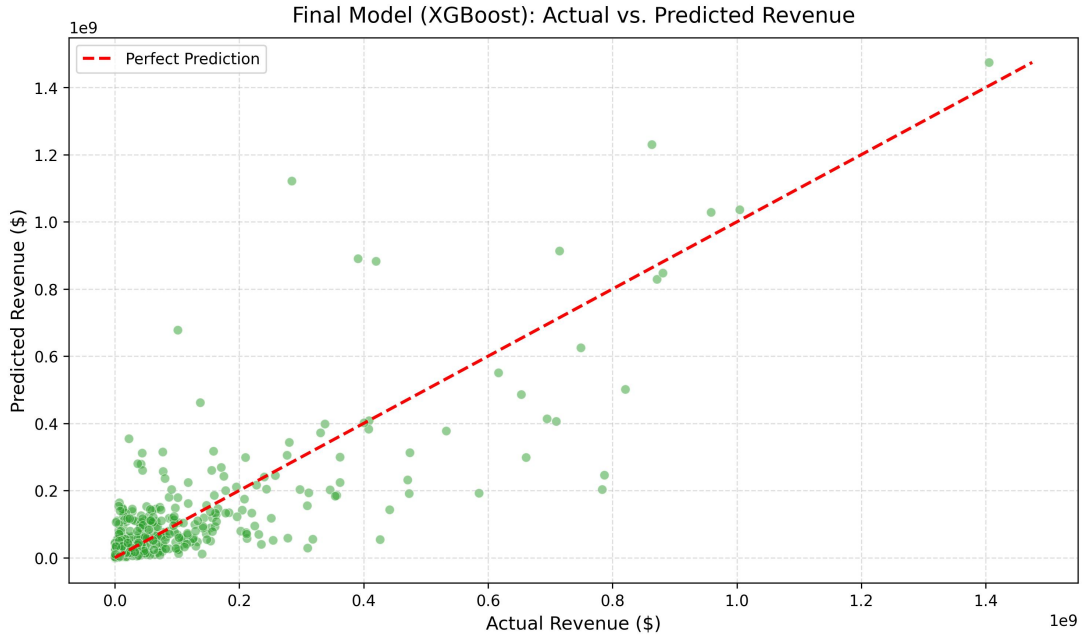


Figure 1: Final Model (XGBoost): Actual vs. Predicted Box Office Revenue.

## 5 Discussion of Challenges

- **The Asymmetric Nature of MAPE on Skewed Data:** During our evaluation, we encountered a significant challenge regarding the requested MAPE metric. Initially, the full dataset yielded a MAPE of over 1,000,000%. We discovered Kaggle artifacts where unknown movie revenues were coded as \$1 or \$5. Because MAPE divides the absolute

error by the actual value, predicting a normal box office for a \$1 placeholder created a mathematically infinite percentage distortion. We resolved this by implementing a strict domain filter (`df['revenue'] > 10000`). While this reduced the MAPE to 2096.76%, it remains heavily skewed. This highlights a fundamental mathematical limitation of MAPE: it penalizes over-predictions on low-revenue films exponentially more than under-predictions on high-revenue films. For example, predicting a \$2M revenue for a limited-release indie film that actually grossed \$50k results in a 3900% error. This drastically skews the average, even though the absolute error (\$1.95M) is insignificant compared to the multi-million dollar scale of blockbusters. Therefore, while MAPE provides context for small relative errors, RMSE remains the more reliable metric for a target variable with such massive scale variance.

- **Missing Feature Adaptability:** Upon migrating to the full Kaggle dataset, we discovered that certain features present in alternative TMDb subsets (such as `vote_average` and `vote_count`) were absent in our specific competition data. We corrected this by engineering temporal features (`release_year`, `release_month`) from the datetime column to compensate for the lost audience metrics, which successfully maintained the robustness of our feature space.
- **High-Cardinality Categorical Features:** Expanding to the full dataset introduced obscure secondary languages into the `original_language` feature. We resolved the subsequent sparse matrix explosion by utilizing a `SimpleImputer` set to the most frequent denominator and ensuring our `OneHotEncoder` handled unknown variables seamlessly during cross-validation (`handle_unknown='ignore'`).

## 6 Conclusion

By leveraging a gradient-boosted architecture and an expanded feature set, we successfully modeled the TMDb Box Office dataset to account for nearly 64% of the variance in global revenues. The project highlights the importance of robust data preprocessing, the mathematical nuances of evaluation metrics like MAPE on highly skewed financial data, and the effectiveness of tree-based ensembles in tabular regression tasks.